Question Paper

Exam Date & Time: 07-Jul-2023 (09:30 AM - 12:30 PM)

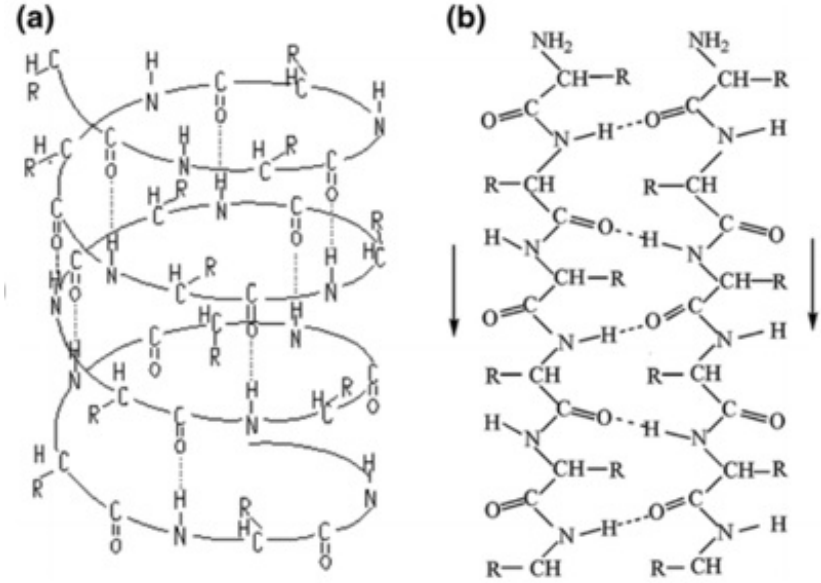
MANIPAL ACADEMY OF HIGHER EDUCATION

SECOND SEMESTER B.TECH. EXAMINATIONS - JUNE/JULY 2023
SUBJECT: BIO 1071-CHM/BIO 1051-B/BIO 1071-B - BIOLOGY FOR ENGINEERS
(MAKEUP)

Marks: 50 **Duration: 180 mins.**

Answer all the questions.

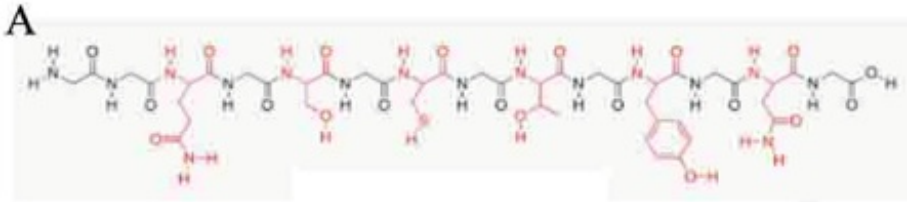
1A) i) What is the stabilizing force behind the structures a and b? Explain. (3)
ii) Which group is not involved in this structure formation? Justify.
iii) What are these structures and why these are different?

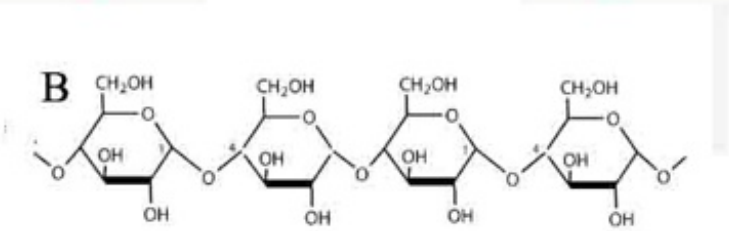


(1+1+1 = 3 marks)

1B) A news statement: "Silicon-Based Life May Be More Than Just Science Fiction. Scientists are showing that nature can evolve to incorporate silicon into carbon-based molecules - the building blocks of life on Earth." (3)
i) How silicon based life is possible?
ii) What is the scientific reasoning behind the second statement of the news?
iii) Why only silicon can incorporate why not aluminium which is abundant on the earth's crust? Justify.
(1+1+1 = 3 marks)

1C) Compare and analyse the two polymers found in living systems. (4)



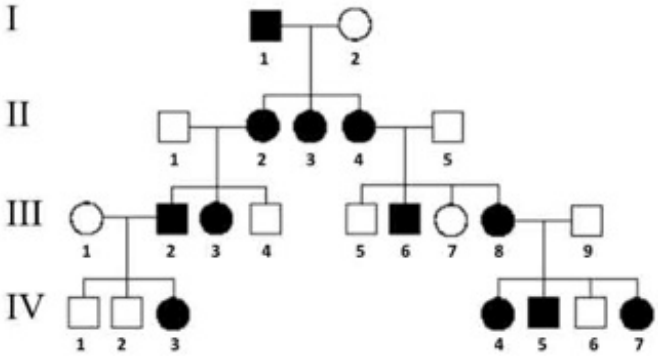


i) What is the "engineering" in these structures making it different?
ii) If you heat which one will lose its function first? Why?
iii) Hydrogen bonding plays a big role in the structural diversity of one of the above main class. How?
(2+1+1 = 4 marks)

2A) You are a teaching assistant for Biology for Engineers. You have pure-breeding wheat plants of a particular kind which produce long and thin seeds (Dominant, LLTT). You also have pure-breeding plants of short and thick seeds (ttll). You asked your student project assistant to cross the two and to give you the results of F1 and F2 crosses. The results F2 were: (3)

Long wheat, thick seeds = 186
Long wheat, thin seeds = 564
Short wheat, thick seeds = 63
Short wheat, thin seeds = 187
i) Whether the factors are on same or different chromosomes? Justify
ii) Construct a Punnett square for F2 generation, what is the expected?
(1+2 = 3 marks)

2B) i) What is the most probable mode of inheritance in the following pedigree? Why? (3)

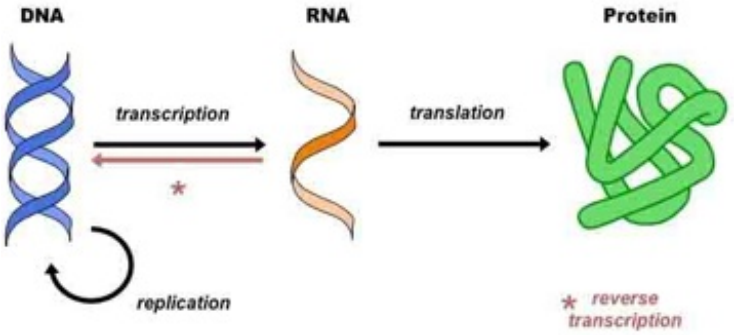


ii) Predict the genotype of the individuals III-2 and IV-3
(2+1 = 3 marks)

2C) In *Drosophila melanogaster*, there is a dominant factor (Allele) for gray body color (B) and another dominant factor for normal wings (V). The recessive alleles of those two genes result in black body color (b) and vestigial wings (v), respectively. Flies homozygous for gray body and normal wings were crossed with flies that had black bodies and vestigial wings. The F1 progeny were then test-crossed, with the following genotypes: (4)
BbVv: 236: bbvv: 253: Bbvv: 50: bbVV: 61

i) By looking into the results would you say that these two genes are linked? If so, then show the above cross.
ii) Mention the genotypes of parental generation. Classify the parental and recombinant genotypes.
iii) Calculate the recombination frequency and Map distance?
(2+1+1 = 4 marks)

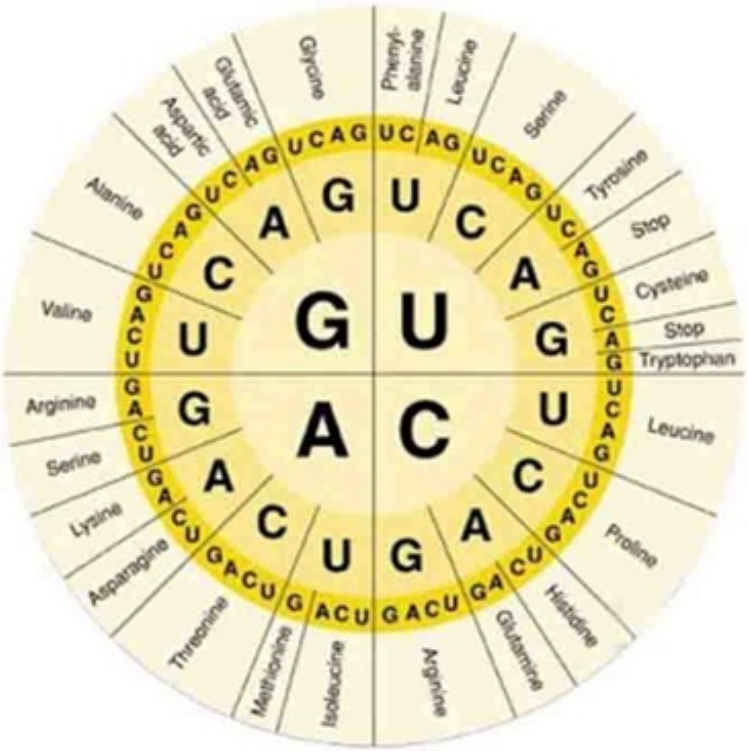
3A) An engineering student has made an illustration regarding information storage and its processing in a cell as follows: (3)



Analyse the figure and suggest logical explanations for each step.

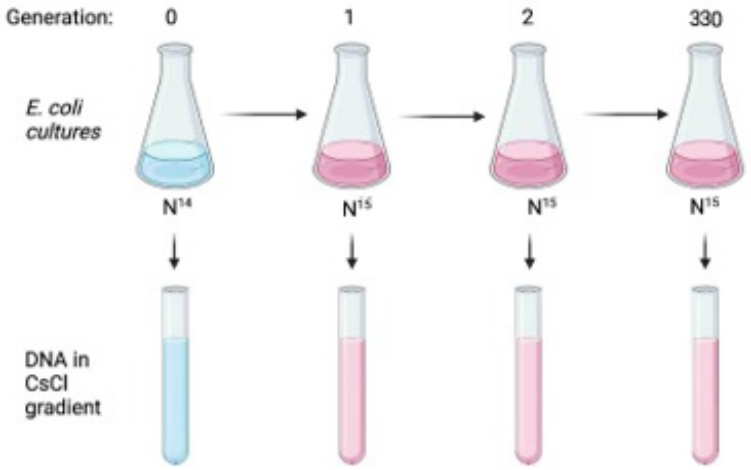
3B) A non-template DNA code 5'TCG3'. Find the corresponding RNA code, anticodon code in tRNA and the amino acid. Label the ends. Provide your answer in the following column format (3)

RNA code with labelled ends	
tRNA anticodon with labelled ends	
Corresponding amino acid	



(1+1+1 = 3 marks)

3C) Given below is a illustration of Meselson and Stahl. The student started with **N^{14} cultures and then shifted to N^{15}**. (4)



i) Copy the figure and mark the band accumulation in each of the generation given.
ii) Calculate the percentage of DNA at the end of 2nd generation bearing N^{15}/N^{15} strand.
iii) Can you rule out dispersive mode and conservative mode of replication using this experiment? Justify.
(2+1+1 = 4 marks)

4A) An animal's DNA contains a gene that codes for a protein that produces a compound required for several essential metabolic reactions. Over time, that organism develops a "broken" form of the gene (called a pseudogene) and is no longer able to produce that compound. Why might it be advantageous for the organism to develop this "broken" gene? (3)

4B) (i) It is said that the harmful alleles get eliminated from the population over time, yet sickle cell anaemia is persisting in the human population. Why? (3)
(ii) A hummingbird species feeds on a flower species with a similarly shape. These flowers also depend on the hummingbirds to pollinate them. Which could be a possible explanation for this?
(2+1 = 3 marks)

4C) Explain the type of species interaction illustrated in the following diagrams (4)

Example	Species interaction

5A) (i) Why do we need Booster dose after vaccination? (3)
(ii) Which type of vaccine is prepared by typically killing the pathogen by using heat or chemicals like formalin?
(iii) When the vaccine virus is given to a human, it will be unable to replicate enough to cause illness but will still provoke an immune response that can protect against future infection. Name the type of vaccine.

5B) The first pertussis vaccines, introduced in the 1940s, comprised inactivated Bordetella pertussis bacteria. Although effective, whole-cell pertussis vaccines frequently caused minor adverse reactions such as fever and swelling at the injection site. This caused many people to avoid the vaccine, and by the 1970s, decreasing vaccination rates had brought about an increase in new infections. Basic research at NIAID and elsewhere, as well as NIAID-supported clinical work, led to the development of acellular (not containing cells) pertussis vaccines that are based on individual, purified B. pertussis components. These vaccines are similarly effective as whole-cell vaccines but much less likely to cause adverse reactions. (3)
i) Although these vaccines are safer and easier to produce but antigens alone are not sufficient to induce adequate long-term immunity. How would you increase the immunogenicity of the vaccine?
ii) What are the advantages of subunit vaccines?
iii) B. pertussis produces toxins which affect host-pathogen interactions can these toxins be used as vaccines, are these toxins safe to use.
iv) Is it possible for us to prevent diseases without vaccines.

5C) Give one example of living organisms which Bioinspired to create the following technology. (4)

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